

4.1 Method Used

Joint Application Design (JAD)

JAD refers to a structured workshop, which involves bringing in the users, stakeholders, and developers into a collaborative environment to discuss the requirements of the system. Unlike individual one-to-one interviews, JAD ensures that all interested parties are involved in one conversation to discuss their needs and expectations for the new system, as well as identifying any existing problems that must be addressed. Hence, a JAD session is appropriate because it allows our team to obtain the perspectives of different stakeholders simultaneously, reduce misunderstandings in the early process, and produce a richer set of requirements in a shorter time compared to conducting separate individual interviews.

JAD Session Details :

Item	Details
Participants	UTM students (representing bus users, carpool users, freshmen)
Session Leader	Chew Jian Hui, Darren Chua Boon Yee
Duration	Approximately 45 minutes
Location	Faculty of Computing, UTM JB
Topic Covered	System scope presentation, validation of proposed modules, user expectations, safety and privacy concerns

Key Discussion Questions that Raised during the JAD session:

1. Based on what was presented, do you think this system would actually solve the transport problems you face daily?
2. What information would you need from the app before deciding whether to wait for or skip a bus?
3. Have you ever used informal carpooling on campus? What made you uncomfortable about it?
4. When navigating to an unfamiliar location on campus, what kind of guidance would you expect from an app?
5. If something went wrong during your journey, how would you want to report it?
6. Are there any concerns you have about using a campus transport app, particularly around privacy or safety?

Structured Observation of the Environment (STROBE)

STROBE is an unobtrusive method, which allows us to observe the user behaviour and how UTM transport works without directly interacting with or influencing anything. Rather than just merely relying on what users report about their experience, STROBE successfully captures what actually happened and provides a first-hand view of the current system shortcomings. In fact, the students may sometimes fail to report an accurate past travel experience during the JAD session or questionnaire. Such failure could result in underestimating the waiting time experienced by the students or failing to articulate the physical conditions they endure at bus stops on a daily basis. In that case, STROBE comes in handy since our team can witness firsthand the true nature of the bus stops, number of students, and behaviour during peak and non-peak periods.

Observation Items	Details
Average number of students waiting at bus stop	Counted at 5-minute intervals
Bus occupancy upon arrival	Empty / Available / Moderate / Full
Number of students unable to board	Counted per bus
Carpool activity in Telegram group	Any students observed arranging rides via phone while waiting
Bus schedule in UTM Smart	The way it shows the bus arrival time and destination

4.2 Summary from Method Used

JAD Session Summary

At the beginning of the session, an overview of the system scope was presented, which gave participants enough context to provide focused and meaningful feedback rather than just simply complaining about the situation.

Participants agreed that the new system will address their issues, but they pointed out that it will greatly depend on the reliability of information that will be displayed in the application. A recurring point was that since the current system only provides a fixed timetable with no real-time updates, students have no way of knowing whether a bus is running on time or how full it already is. Many participants emphasized the severity of this issue, because they frequently missed the morning classes or were left stranded as full buses drove past them without stopping. Participants stressed that for the new app to be trusted, its live data must be consistently accurate, otherwise it would be no more reliable than the static schedule they already have.

On the question of what information they needed before deciding whether to wait for a bus, participants consistently highlighted three things, which are current bus location, estimated arrival time, and current occupancy level. Some participants mentioned that the occupancy level was even more important than the time of arrival.

The carpooling discussion showed that most participants were familiar with the Student Grab via Telegram group but felt uncomfortable and awkward using it due to the lack of identity verification. They agreed that contact details should only be shared after both parties confirm a match, so that restricting the platform to verified UTM accounts was a necessary safety measure rather than just a nice-to-have.

Freshmen participants raised the route navigation issue most strongly, pointing out their difficulties in understanding which bus lines to use and where to make transfers when visiting unfamiliar places of campus because UTM Smart's current approach of redirecting to Google Maps was unhelpful as it does not consider any bus routes within the campus network at all. They agreed that a useful route planner must incorporate available bus routes and transfer points into its suggestions.

Regarding the feedback mechanism, participants expressed great dissatisfaction with the lack of official ways to report repeated issues, such as bus lateness and misconduct by drivers and these same problems had persisted for many semesters without noticeable actions from the university admin. To solve this, instead of a lengthy and tedious feedback form that most students would typically ignore, participants were supportive of a post-trip rating prompt with categorised complaint options, because most students would not bother with a lengthy feedback form but would readily tap a category and a star rating.

STROBE Summary

We successfully conducted the direct observation in several UTM bus stops and finally obtained the following findings.

During the morning peak hours, especially 7:15 am until 8:00 am, average 20 to 25 students were observed waiting at the bus stop at any given 5 minute interval. Among the buses that arrived during the observation period, approximately 60% arrived already at moderate-to-full capacity, leaving a significant number of waiting students unable to board. Many students were seen to board only after waiting for several consecutive buses, with waiting times exceeding 25 minutes in some cases. The same bus route was consistently observed arriving full across multiple consecutive arrivals.

Moreover, although UTM offers a bus schedule feature in UTM Smart where students can choose any bus and see its complete timetable, which shows the departure time, starting point, and the final point of the bus, this level of information is far from being enough for practical usage. However, the estimated time for intermediate stations are completely missing from the schedule, leaving only a vague time frame within which the bus should pass the stop. Additionally, there are no intelligent searches offered to the users, which means that it is impossible to find the best bus according to any particular point of interest such as a certain time or destination. In UTM Smart, we also observed that there is a feature to show some common locations in the UTM campus, but it just directly uses the Google map to show the route without specifying the way to utilize the free buses provided by UTM to go there.

Besides that, we also observed that there exists an informal carpooling activity via the Telegram group. While no dedicated platform exists for this purpose, it is evident that students have organically formed their own workaround to coordinate rides among themselves. However, this approach remains fundamentally limited, as there is no formal verification of driver identities, no structured matching system, and no accountability mechanism, leaving both drivers and passengers without adequate safety assurance.